Summary SCIENTIFIC REPORT

Contract No. 63 TE / 08.01.2025

Project title: Modelling Uncertainty and Spatial Effects in Time – Applications in Finance

Project code: **PN-IV-P2-2.1-TE-2023-1317**

Project duration: 24 months

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Scientific description

The MUSET project – **Modelling Uncertainty and Spatial Effects in Time –** Applications in **Finance** – focused primarily on conducting theoretical and empirical investigations and strengthening the methodological foundations. As set out in the research proposal, the main objective is to model uncertainty in financial data and spatial dependence, not only by applying existing methodologies but also by developing new ones. This is driven by the aim of reducing uncertainty. At the same time, a degree of uncertainty persists when addressing various issues, and this uncertainty is conditioned by individual, temporal, and spatial processes.

Summary of objectives

Main objectives	
1. RD1. Methodological development of a new CUBT approach RD1.1. Modelling uncertainty in different types of ordinal data obtained from financial surveys. RD1.2. Methodological development of CUBT – model development. RD1.3. Methodological development of CUBT – statistical inference. RD1.4. Methodological development of CUBT – model validation. RD1.5. Software implementation of CUBT – development of the CUBT R package.	✓

2. RD2. Modelling uncertainty and spatial effects over time RD2.1. Applying the new CUBT approach to a new sample of longitudinal financial data. RD2.2. Developing generalised spatial and spatio-temporal ARCH models for cryptocurrencies and comparing the results with those obtained for traditional stock markets and the banking sector. RD2.3. Developing generalised spatial and spatio-temporal ARCH models for companies' financial data.	✓

In addition to these research directions, secondary objectives have also been defined to ensure open access to the project results:

Secondary objectives	
● SO1. Creating a website and social media accounts for the project to present its results. SO2. Providing the public with free CUBT-related materials, such as a user guide. SO3. Disseminating the results in academia and the private sector. ● SO2. Punerea la dispoziția publicului, în mod gratuit, a unor materiale dedicate CUBT, precum un ghid de utilizare. ● SO3. Diseminarea rezultatelor în mediul academic și în sectorul privat.	✓

Summary of deliverables

Deliverables	**Proposed number**	**Level of achievement**
Articles	3 articles in WoS Q1 or Q2 journals	100% (3 articles published)
	3 WoS- or Scopus-indexed articles	133% (4 articles published)
	–	More than 5 articles submitted for publication
Conferences/events evenimente	At least 10 events, with at least one participation per team member per year cu cel puțin o participare per membru al echipei pe an.	Year 1: 18; Year 2: 9. Achievement rate: 270% (27 conferences, excluding other events). Each team member participated in at least one conference per year. Grad de realizare: 270%, (27 de conferinţe, fără alte evenimente) fiecare membru a participat la cel puţin o conferinţă pe an
Book chapter	1	100% (1 chapter)
R application	1	200% (2 applications)
Website	1	100% (1 website)
Social media	2	100% (2 LinkedIn and Facebook pages)
GitHub project	1	400% (4 GitHub projects)
Quantinar platform guide	1	100% (1 guide)
Podcast	3	233% (7 podcasts)

All objectives and results envisaged in the project implementation plan have been achieved. The deliverables are as follows:

a) Published scientific articles

Year 2025

1. Danci, A. R., Lazar, I. A., & **Mureșan, G. M.** (2025). Financial analysis in the era of uncertainty: a comprehensive examination. Digital Finance, 1-25.

Summary: This study presents an extensive bibliometric analysis of the evolution of research in financial analysis over the period 1992–2024, in a context characterised by uncertainty, digitalisation, and the transition towards sustainable finance. Based on a dataset extracted from the Web of Science and using Python and VOSviewer, more than 4,000 relevant articles were identified. The results show a significant increase in academic interest in the topic and the rapid evolution of financial analysis towards an interdisciplinary approach. The conclusions identify the integration of ESG factors into financial models, the impact of artificial intelligence, and the modelling of uncertainty in finance as future research directions.

2. **Achim, M. A**., & **Mare, C.** (2025). Factors influencing the shadow economy: an international analysis. Journal of Risk Finance, 1–17.

Summary: This study examines the determinants of the shadow economy in the 27 European Union Member States over the period 2013–2022, using a spatial panel Spatial Durbin Model (SDM). It is based on the premise that a high tax burden and poor governance quality generate concerns and uncertainty among taxpayers, thereby increasing the intention to evade taxes and reducing budget revenues. The results show that the shadow economy is negatively associated with budget revenues, population growth, regulatory quality, and economic growth, with significant spatial spillover effects. The rule of law and the level of economic development are among the most important determinants. Reducing the shadow economy decreases uncertainty surrounding budget revenues and enables better planning of their use.

3. **Mureșan, G. M.,** Fülöp, M. T., & Cifuentes-Faura, J. (2025). Does trust influence policyholder intent and behavior? Evidence from private health insurance. International Journal of Healthcare Management, 1–14.

Summary: This study extends the Theory of Planned Behaviour (TPB) by incorporating trust as an antecedent of intention and online purchasing behaviour in private health insurance. The data come from an online questionnaire completed by 1,300 respondents in Romania and are analysed using PLS-SEM. Perceived behavioural control is the strongest predictor of intention ($\beta = 0.462$), followed by trust ($\beta = 0.231$), attitude ($\beta = 0.152$), and subjective norms ($\beta = 0.098$). Trust also significantly influences actual behaviour ($\beta = 0.211$), while intention largely explains behaviour ($\beta = 0.594$; $R^2 = 0.779$). In the digital environment, trust in the insurer and the perceived ability to independently assess and complete an insurance policy matter more than social influence.

Year 2026

4. Spelta, A., Facchinetti, S., Osmetti, S.A., Tarantola, C., **Mare, C.,** Mazzali, A., Merli, L., Iannario, M. (2026). A network-based distributional inference approach to model country-level cyber risk.

Journal of the Royal Statistical Society Series A: Statistics in Society;, qnag049, https://doi.org/10.1093/jrsssa/qnag049

Summary: The article proposes a network-based distributional inference approach to modelling cyber risk at country level. The dataset comprises 7,059 daily cyberattacks recorded in 2023 and provided by Hackmanac, combined with macroeconomic indicators to construct country profiles. The distributional network and inference are implemented through Wasserstein Propagation in Social Networks. The results highlight interdependencies and spatial vulnerabilities between countries and demonstrate the importance of combining network and spatial analyses with uncertainty-modelling methods to inform strategies aimed at reducing losses caused by cyberattacks.

5. Ozturkkal, B., Zdravevski, E., Tanda, A., **Mare, C.,** Wahlstrom, R. R., Lameski, P., Trajkovik, V. (2026). Perspectives on venture capital: A practitioner-oriented bibliometric and systematic review. Open Research Europe , 6:162 (https://doi.org/10.12688/openreseurope.23068.1)

Summary: This study provides a practitioner-oriented bibliometric and systematic review of the literature on venture capital financing. A total of 1,264 papers published between 2010 and 2022 were selected from five databases, using a natural language processing (NLP) toolkit based on keywords and covering three thematic areas: ESG, corporate social responsibility and sustainability, innovation, and exit strategies. The results show a growing body of research, largely based on traditional methods and focused on the US and Chinese markets, indicating a gap in the literature on emerging markets. The analysis confirms that innovation is essential to venture capital investment performance.

6. Moroșan, A., **Belbe, Ș.,** Mihaiu, D. M., Bunescu, L., Vasiu, D. E., & Tăvală, F. M. (2026). Exploring the nexus between environmental taxation, air pollution, and health outcomes in emerging and non-emerging EU countries: a space-time perspective. Frontiers in Public Health, 14, 1896504.

Summary: This study investigates the mechanism through which environmental taxes indirectly improve public health by reducing pollution, using a dynamic panel of the 27 EU Member States for 2008–2024. Spatial SLX and Durbin (SDM) models are estimated and compared, distinguishing between emerging economies (Romania, Bulgaria, Hungary, and Poland) and the rest of the EU. Pollution and resource taxes have the strongest pollution-reducing effects, which are also felt in neighbouring countries, indicating significant spatial externalities, whereas energy and transport taxes have limited or no effect. The results support differentiated but coordinated EU-level policies, given the transboundary nature of pollution.

7. **Achim, M. A., Belbe, Ș., Mare, C**., & Otto, P. (2026). Spatial Effects and Uncertainty in Cryptocurrencies–the case of Network ARCH Models. Finance Research Letters, 110729.

Summary: This study analyses cryptocurrency volatility using a balanced daily panel of 50 cryptocurrencies covering 1,517 trading days (06 January 2021–02 March 2025), with neighbourhood matrices based on return correlations (correlation threshold $|r| > 0.15$ and $k = 5$ nearest neighbours). Estimation is carried out in two stages: a spatio-temporal ARMA filter for the mean and a dynamic spatial ARCH model estimated by quasi-maximum likelihood. The results show strong contemporaneous transmission of price volatility ($\rho \approx 0.43$) and an important spatio-temporal effect ($\delta \approx 0.34$), whereas volume volatility is largely idiosyncratic ($\rho \approx 0.12$).

Diversification within the market offers limited protection, and systemic risk should be assessed at the network level rather than for individual coins.

b) Scientific articles submitted for publication

1. **Beleiu, I.N., Achim, M.-A., Mare, C.,** Mureșan, **G.M.** CUB Models in Ordinal Choice Modelling: A Bibliometric and Conceptual Review of Preference and Response Uncertainty. Submitted to Research in International Business and Finance.

Summary: This article is a bibliometric and conceptual review of CUB/CUP models, focusing on two dimensions: their methodological development and evolution over time, and their applications. The bibliometric analysis includes 98 specialised articles indexed in the Web of Science and identifies trends, leading researchers, and the journals making the greatest contributions. It also presents the evolution of statistical methods for ordinal data associated with CUB models. The conclusions propose research directions for analysing uncertainty and sentiment using CUB methods, as well as extending the approach to CUP methods.

2. Raddi, F., **Mare, C.** The Effects of Annual Reports on Volatility: A European Point of View. British Accounting Review.

Summary: This study explores how volatility, viewed as a factor of uncertainty, is affected by annual and sustainability reports, which aim to reduce uncertainty and demonstrate companies' engagement in the market. The results provide a basis for applying spatial volatility methods (SARCH, SGARCH) in the project's second stage. The report provides no further details on the results.

3. Stanca, L., Belbe, S., Gomez, L., Wenzlaff, K., Mare, C. *AI in Quantifying Sustainability Impact on Crowdfunding Outcomes: Evidence from SDG-Aligned Campaigns*. International Review of Economics and Finance.

Summary: This study analyses uncertainty in the crowdfunding market for campaigns aligned with the Sustainable Development Goals, using artificial intelligence. It shows how sustainability-related aspects influence the probability of campaign success and support venture capital (VC) funds in making better-informed investment decisions. Integrating sustainability indicators contributes to efficiency and uncertainty reduction.

4. **Mare, C.,** Stanca, L. Dynamic CUB Models for Ordinal Panel Data – CUBP: An R Shiny Application with Location-Scale and Location-Shift Extensions. Jurnal

Summary: This article extends the CUB model to ordinal panel data, distinguishing genuine preference from decision-making uncertainty. Sentiment and uncertainty are modelled through logit equations with individual random effects and autoregressive AR(1) processes, each with its own persistence coefficient. Location-scale and location-shift extensions are included, and estimation is performed using the EM algorithm with adaptive Gauss–Hermite quadrature. The model is implemented in an R/Shiny application and tested on a synthetic panel of 30 respondents across three waves; this is a software test, not an empirical study.

5. Horga, I., **Mare, C.** Modeling Attitudes and Perceptions from Ordinal and Textual Data: A CUB–Sentiment Approach.

Summary: This article integrates ordinal responses and open-ended questionnaire responses within a single framework. Its novel contribution is the inclusion of sentiment extracted from text, measured using a normalised indicator of positive and negative responses, as an explanatory variable for the CUB parameters. For 16 of the 18 items, uncertainty does not exceed 0.22, while sentiment is high (0.66–0.87). The sentiment indicator, validated against manual coding (Cohen's $\kappa = 0.71$), is significantly correlated with ratings (Spearman's $\rho = 0.29$–0.36). Text does not replace ordinal ratings, but it can help pinpoint issues indicated by the scale through uncertainty and low sentiment.

6. **Mureșan, G.M., Mare, C**. What makes research impactful? Insights from Highly Cited Papers. International Review of Financial Analysis.

Summary: This study analyses 162 highly cited articles (the top 1% in the Web of Science) from the International Review of Financial Analysis using bibliometric mapping, co-citation analysis, and thematic synthesis. Scientific impact has been particularly strong since 2023, with an emphasis on digital transformation, ESG, and climate-related financial risks. The literature is predominantly quantitative (panel data, Difference-in-Differences, Propensity Score Matching, and instrumental variables) and is organised into eight clusters. Artificial intelligence, digital finance, and sustainability are priority directions for high-impact research.

7. **Mare, C., Achim, M.-A., Mureșan, G.M.** Effects in Local Productivity Volatility: A Non-Dynamic Spatial ARCH Analysis of Romanian Localities. Empirical Economics.

Summary: This study extends spatial ARCH models from financial markets to the real economy. More than 9 million records for local administrative units (2008–2023) are aggregated, and labour productivity (turnover per employee) is filtered for locality and year fixed effects. The non-dynamic spatial ARCH model is estimated separately for each year, using the five nearest neighbours to define neighbourhoods. Spatial dependence in conditional variance is confirmed in 7 of the 16 years ($\hat{\rho}$ between 0.14 and 0.27), during periods of aggregate disruption (crisis, pandemic, inflation). Exposure to aggregate shocks has a locational component, which is relevant for regional policy and risk assessment by financial institutions.

8. Iannario, M., **Mare, C., Mureșan, G.-M., Vaidean, V. L., & Achim, M. V. (**2026). *Modelling uncertainty in happiness sentiment perception during the COVID-19 pandemic using statistical models accounting for heterogeneity.*

Summary: This study applies CUB and CUP models to a sample of 1,572 respondents from 43 countries, surveyed at three time points about their perceptions of happiness amid financial shocks during the COVID-19 pandemic. The most effective specification is the model with location and scale parameters (BIC = 3455.939, compared with 3563.672 for the proportional-odds model). Income fluctuations increase instability and uncertainty regarding happiness. Men and older individuals are more stable, whereas women and younger individuals are more vulnerable.

c) Published book chapters

1) Danci, A. R., Lazar, I. A., & **Mureșan, G. M. (2026).** One More Time, the Dark Side of the Economy: Exploring the Shadow Economy in the European Union. In Financial Crime in the Digital Era: Innovative Approaches to Preventing Financial Fraud (pp. 445-468). Cham: Springer Nature Switzerland.

Summary: This chapter analyses the determinants of the shadow economy in the European Union over the period 2003–2022, using panel-data analysis techniques and socioeconomic, cultural, and institutional variables. It draws on the indicator developed by Friedrich Schneider. The results show that cultural, institutional, and behavioural factors significantly influence the size of the shadow economy, while effective governance systems can mitigate the effects of tax pressure. Legal and fiscal reforms must be complemented by changes in citizens' behaviour and civic engagement to combat financial crime.

c) Website şi social media

- **Website**: https://muset-te.github.io/ .
- **Facebook :** https://www.facebook.com/profile.php?id=61572720399792, 529299510272114 ID Pagină
- **Linkedin:** https://www.linkedin.com/company/muset-research-team
- **GitHub Organization**: https://github.com/MUSET-TE

d) GitHub Projects

1. **AI in Crowdfunding**
https://github.com/stefanacioban/ai_in_crowdfunding
2. **Fraud Prevention Crowdfunding,**
https://github.com/cost19130/COST_CA19130_fraud_prevention_crowdfunding_p2p_ico
3. **Space-Time ARCH Models**
https://github.com/MUSET-TE/spatial_crypto
4. **CUB/CUP Time Models**
https://github.com/MUSET-TE/cub_cup_time

e) Podcasts

1. **Exploration of CUB Tools.** *Exploration of the CUB Models for modelling uncertainty and their relevance to the MUSET-TE project.*
https://muset-te.github.io/assets/podcasts/Exploration_of_CUB_Tools.mp3
2. **Stop Misleading Data.** *How CUB Models untangle preference from uncertainty in rating scales.*
https://muset-te.github.io/assets/podcasts/Stop_Misleading_Data!_How_CUB_Models_Untangle_Preference_from_U.m4a
3. **Modelling Uncertainty.** *When you make a choice, what's really going on?*
https://muset-te.github.io/assets/podcasts/Modeling%20Uncertainty.mp4
4. **The invisible web.** Connecting our global economies to our digital risks.
https://muset-te.github.io/assets/podcasts/Te%20invisible%20web.mp4
5. **Beyond the stars.** *The hidden psychology of our everyday rating.*
https://muset-te.github.io/assets/podcasts/Beyond%20the%20stars.mp4
6. **Cyber risk. Mapping cyber risk in data blind spots.**
https://muset-te.github.io/assets/podcasts/Mapping_cyber_risk_in_data_blind_spots.m4a
7. **Network-Based Cyber Risk. How network based models are changing the way we understand cyber risk at country level.**
https://muset-te.github.io/assets/podcasts/NoteGPT_AI_Podcast_Network-Based%20Cyber%20Risk%20Propagation.mp3

f) Dissemination of results:

The project team participated in numerous scientific events, as follows:

Year 2025

1. **Achim, M.-A., Belbe, Ș., Beleiu, I.N.**, Otto, P., **Mare, C.** (November 27-29, 2025). *Spatial Effects and Uncertainty in Cryptocurrencies – the case of Space-Time ARCH Models.* 9th CERS - Central European Conference in Regional Science. Sustainability-driven territorial development. Challenges for CEECs, Cluj-Napoca, Romania. **https://cers.rrsa.ro**
2. Xu, Y., **Mare, C.,** Jiang, W., Zhang, D. (November 27-29, 2025). *High Dimensional Threshold Spatial Autoregressive Models: Estimation and Variable Selection.* 9th CERS - Central European Conference in Regional Science. Sustainability-driven territorial development. Challenges for CEECs, Cluj-Napoca, Romania. **https://cers.rrsa.ro**
3. Raddi, F., **Mare, C.** (November 27-29, 2025). *A spectral guide to the spatial weight matrix selection.* 9th CERS - Central European Conference in Regional Science. Sustainability-driven territorial development. Challenges for CEECs, Cluj-Napoca, Romania. **https://cers.rrsa.ro**
4. **Achim, M.-A., Mare, C., Beleiu, I.N., Belbe, Ș.** (November 21-22, 2025). *Trends and Insights in CUB Model Research: A Bibliometric Study Based on WoS.* The 18th International Conference on Applied Statistics, Predeal, Romania. **https://simpstat.ase.ro**
5. Purcel, A., **Mureșan, G.** (November 14-15, 2025). *Developmental trade-offs and the pursuit of happiness: income, emissions, and financial connectivity.* Cluj Economics and Business Conference, Global economy under the auspices of the sustainable development, Cluj-Napoca, Romania. **https://econ.ubbcluj.ro/cebc/programme.php**
6. **Belbe, Ș., Gere, I., Mare, C.** (November 4-7, 2025). *Does Spatial Clustering and Dispersion persist?* Smart Diaspora 2025, Cluj-Napoca, Romania. **https://diaspora-stiintifica.ro/**
7. **Mare, C.,** Tarantola, C., Facchinetti, S., Iannario, M., Mazzali, A., Merli, L., Osmetti, S., Spelta, A. (November 4-7, 2025). *Network Analysis and Uncertainty in Finance: A Statistical Perspective from Our Research on a Cyber Risk Case Study.* Smart Diaspora 2025, Cluj-Napoca, Romania. **https://diaspora-stiintifica.ro/**
8. **Mureșan, G., Mare, C.** (October 8-11, 2025). *A systematic literature review on uncertainty and risk in Finance: theoretical foundations and empirical trends.* Kautz Conference on Business and Economics, Győr, Hungary. **https://kautzconference.sze.hu**
9. **Mureșan, G.**, Sabadâș, G. (May 9-10, 2025). *Literature Review of Uncertainty Risk in Finance.* International Yildirim Bayezid Scientific Research and Innovation Symposium-I, Bursa, Türkiye. **https://www.iksadkongre.com/bursa**
10. **Belbe, Ș.,** Gere, I., **Mare, C.** (May 2-6, 2025). *The Impact of Spatial Aggregation upon unveiling Hidden Spatial Autocorrelation.* The 19th World Conference of the Spatial Econometrics Association & The Spatial Econometrics Advanced Institute, Rabat, Morocco. **https://sites.google.com/site/jefconference/**
11. Lazăr, I., Danci, A., **Mureșan, G.** (March 20-21, 2025). *Is time for a financial EKG - behavior and performance under conditions of uncertainty.* Conferința Europeană a Serviciilor Financiare ECFS 2025, Brașov, Romania. **https://www.isfin.ro**
12. Lazăr, I., Danci, A., **Mureșan, G.** (March 15-16, 2025). *The Financial EKG of an Entity. Behavior and Performance under Conditions of Uncertainty.* IBANESS Congresses Series on Economics, Business and Management, Plovdiv, Bulgaria. **https://www.ibaness.org/index.php**

13. Lazăr, I., Danci, A., **Mureșan, G.** (March 15-16, 2025). *The Relationship Between Financial Manipulation and The Bankruptcy Risk of a Company.* IBANESS Congresses Series on Economics, Business and Management, Plovdiv, Bulgaria. **https://www.ibaness.org/index.php**

14. **Mureșan, G.**, Colța, A., Sabadâș, G. (March 15-16, 2025). *Literature Review of Uncertainty Risk in Finance.* IBANESS Congresses Series on Economics, Business and Management, Plovdiv, Bulgaria. **https://www.ibaness.org/index.php**

15. Lazăr, I., Danci, A., **Mureșan, G.** (February 5-9, 2025). *The Financial EKG of an Entity. Behavior and Performance under Conditions of Uncertainty.* 11th International Paris Congress on Social Sciences & Humanities, Paris, France. **https://www.iksadparis.org/social**

16. Lazăr, I., Danci, A., **Mureșan, G.** (February 5-9, 2025). *The Relationship Between Financial Manipulation and The Bankruptcy Risk of a Company.* 11th International Paris Congress on Social Sciences Humanities, Paris, France. **https://www.iksadparis.org/social**

17. **Mureșan, G.** M., Colța, A. (February 5-9, 2025). *A Bibliometric Analysis Of Uncertainty Risk In Finance.* 11th International Paris Congress on Social Sciences Humanities, Paris, France. **https://www.iksadparis.org/social**

Year 2026

1. **Mare, C.** (March 31, 2026). *Spatial Effects and Uncertainty in Cryptocurrencies – The Case of Space-Time ARCH Models.* Scientific Seminar of the Economics Department, Ca' Foscari University, Venice, Italy.

2. Raddi, F., **Mare, C.** (April 23–25, 2026). *The Effects of Annual Reports on Volatility: A European Point of View.* 8th Annual COMPTEXT Conference 2026, University of Birmingham, Birmingham, United Kingdom. https://www.comptextconference.org/8th-annual-conference-2026/

3. **Mureșan, G. M.** (May 28–31, 2026). *Analiza economico-financiară în era AI – noul rol al profesorului.* A 50-a Întâlnire a Cadrelor Didactice și de Cercetare care Predau Disciplinele de Analiză și Evaluare Economico-Financiară, Bucharest, Romania. https://www.sraef.ro/

4. **Mare, C.** (June 3, 2026). *A Network-Based Distributional Inference Approach to Model Country-Level Cyber Risk.* Scientific Seminar of the University of Edinburgh Business School, Edinburgh, United Kingdom. https://www.linkedin.com/posts/uoebusiness_were-pleased-to-welcome-codruta-mare-to-activity-7467885306420232192-uhy2/

5. **Achim, M.-A., Belbe, Ș., Mare, C.,** Otto, P. (June 17–18, 2026). *Spatial Effects and Uncertainty in Cryptocurrencies – The Case of Network ARCH Model.* 20th World Conference of the Spatial Econometrics Association & 24th International Workshop in Spatial Econometrics and Statistics (SEA & SEW 2026), Paris, France.

6. Raddi, F., **Mare, C.** (July 29–31, 2026). *A Spectral Guide to the Spatial Weight Matrix Selection.* Annual Scientific Conference of Romanian Academic Economists from Abroad (ERMAS 2026), Romania. https://www.ermas.ro/2026/index.php

7. **Belbe, Ș.** (August 25–28, 2026). HiTEc Summer Course Athens 2026. Athens, Greece. https://www.hitecaction.org/athens2026course.php

8. Horga, I., **Mare, C.** Modeling Attitudes and Perceptions from Ordinal and Textual Data: A CUB–Sentiment Approach. (October 22–24, 2026). Globalization and Higher Education in Economics and Business Administration (GEBA 2026), Iași, Romania. https://www.feaa.uaic.ro/geba/

9. **Achim, M.-A., Belbe, Ș., Mare, C.,** Otto, P. (October 22–24, 2026). Spatial Effects and Uncertainty in Cryptocurrencies – The Case of Network ARCH Model. Globalization and Higher Education in Economics and Business Administration (GEBA 2026), Iași, Romania. https://www.feaa.uaic.ro/geba/

In conclusion, the objectives and activities undertaken have not only been fully achieved.

Project Director,
Professor Codruța MARE